

Applying NeuralODE-based Cardiovascular Model Identification for Experimental Data Analysis

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Abstract—Recent model-based diagnostic methods have been found promising to provide non-invasive perfusion markers to assess the efficacy of fluid therapy, the most common treatment method for acute circulatory failure (ACF). The development of these model-based diagnostic methods requires the identification of the central arterial pressure curve based on the femoral arterial pressure. This current study presents improvements of the previously suggested NeuralODE-based identification method by suggesting the use of a physiologically interpretable parameter set of the Tube-load model-based transfer function for the physiological system analysis and suggesting a calculation method decreasing the measurement error-caused uncertainty of the identification parameter, called pulse transfer time.

Index Terms—model-based medical therapy, cardiovascular diagnostics, physiological system modelling, parameter identification, transfer function, NeuralODE, intensive care, personalized medical therapy

I. INTRODUCTION

In the case of Acute Circulatory Failure (ACF), fluid therapy is a widely used intervention to stabilize heart function [1]. However, the effectiveness of fluid therapy is not directly predictable, and the therapy can also be harmful. Physiological models can predict fluid responsiveness - describing the effectiveness of fluid therapy for the patient -, but aortic (central arterial) blood pressure must be used to identify their parameters [2]–[5]. The measurement of central arterial

blood pressure is highly invasive and uncommon in clinical practice. However, the aortic pressure can be approximated using femoral arterial pressure measurements ([6], [7]). The Tube-load model [8], [9] describes the relationship between the femoral and central arterial blood pressures if the correct parameters of the model are known for the current patient. In our previous study, a NeuralODE-based parameter identification method for the transfer function parameters was suggested [10]. By identifying the transfer function parameters for a given physiological measurement, the parameter values can be examined, and their connection to different physiological states can be defined.

The aims of the current study are twofold. At first, two system parameters derived from the transfer function parameters are proposed that are more appropriate for describing the patient's blood vessels' state. The second objective is to suggest a calculation method that decreases the measurement error-caused uncertainty of the identification parameter, called pulse transfer time.

II. METHODS AND DATA

A. Experimental Data Set

The experimental data set used in this study was collected from experiments with 8 pure Piétrain pigs performed at the Central Hospital of the University of Liege (Reference Number 14-1726). The pigs underwent different haemodynamic interventions during the measurements, as it is shown in Figure 1. The pigs were mechanically ventilated, and a number of respiratory recruitment manoeuvres were performed in which positive end-expiratory pressure (PEEP) is increased in steps of 5 cmH₂O. Increasing PEEP reduces systemic venous return

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to the right heart and increases pulmonary resistance. Further details of the data collection can be found in [10].

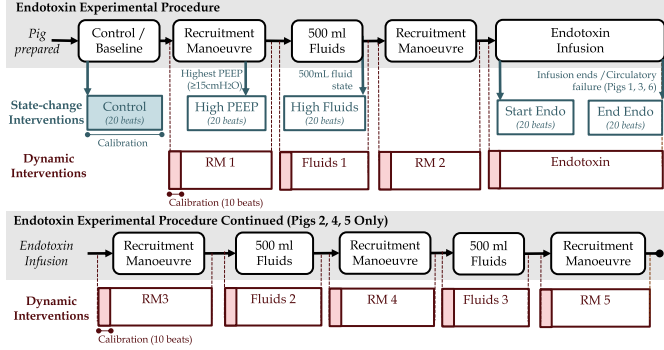


Fig. 1. The schedule of interventions performed on the examined pigs [11].

B. Transfer function derived from the Tube Load model

The Tube load model describes the cardiovascular system as parallel lossless tubes with loads at the end of the tubes. In this study, a simple zero-pole model is used as the load, described with parameters A and B as follows:

$$Z_{load}(\omega) = Z_c \frac{j\omega + B}{j\omega + A}, \quad (1)$$

where $(0 < A < B)$ and Z_c are the impedance of the corresponding tube, ensuring pressure wave reflections are positive, stable, and become negligible at high frequencies. As the frequency becomes very large, the peripheral load tends to Z_c , and the tube and load are impedance matched, so there are no reflections. The model can be used to create transfer functions between central and peripheral pressures and flows in the frequency domain. Using the Laplace transform, the transfer function can also be represented in the continuous time domain:

$$\begin{aligned} \dot{P}_{ao,est}(t) + BP_{ao,est} &= \dot{P}_{fem}(t + PTT) \\ &+ \frac{B+A}{2}P_{fem}(t + PTT) + \frac{B-A}{2}P_{fem}(t - PTT), \end{aligned} \quad (2)$$

where $P_{ao,est}$ is the estimated central arterial blood pressure, P_{fem} is the measured femoral arterial blood pressure, and PTT is the pulse transit time - the time it takes the blood flow to reach to the femoral arterial measurement site from the central arterial measurement site.

Equation (2) describes the transfer function with parameters A, B and PTT , the parameters that are identified by the NeuralODE-based identification method.

C. Deriving model parameters with physiological meaning

Using the Tube Load model, the reflection coefficient of the aorta (Γ) is given as:

$$\Gamma = \frac{Z_L(j\omega) - Z_c}{Z_L(j\omega) + Z_c}, \quad (3)$$

where Z_L is the frequency-dependent load impedance - also called input impedance -, Z_c is the characteristic impedance

of the tube, and Γ is the reflection coefficient defining the ratio of reflected pressure (P_r) to forward travelling pressure (P_f) waves at point 0 (i.e. the point of reflection): $P_r(0, j\omega) = P_f(j\omega) \cdot \Gamma(j\omega)$. The details of the Tube Load model can be found in the PhD thesis of Liam Murphy [12].

The propagation of blood pressure waves can be expressed as the sum of forward travelling and reflected waves:

$$P_f(x, j\omega) = P_f(0, j\omega)e^{j\omega PTT \frac{x}{d}} \quad (4)$$

$$P_r(x, j\omega) = P_r(0, j\omega)e^{-j\omega PTT \frac{x}{d}} \quad (5)$$

$$\begin{aligned} P &= P_f(0, j\omega)e^{j\omega PTT \frac{x}{d}} + P_r(0, j\omega)e^{-j\omega PTT \frac{x}{d}} = \\ &P_f(0, j\omega) \left[e^{j\omega PTT \frac{x}{d}} + \Gamma(j\omega)e^{-j\omega PTT \frac{x}{d}} \right], \end{aligned} \quad (6)$$

where d is the length of the tube, x is the distance from the endpoint of the actual position in the tube, and $PTT \frac{x}{d} = t$.

The central arterial (aortic) and peripheral arterial blood pressures are the pressures at the beginning of the tube and at the point of reflection:

$$P_{ao}(j\omega) = P_f(0, j\omega) [e^{j\omega PTT} + \Gamma(j\omega)e^{-j\omega PTT}] \quad (7)$$

$$P_{fem}(j\omega) = P_f(0, j\omega) [1 + \Gamma(j\omega)] \quad (8)$$

$$\frac{P_{ao}}{P_{fem}} = \frac{e^{j\omega PTT} + \Gamma(j\omega)e^{-j\omega PTT}}{1 + \Gamma(j\omega)} \quad (9)$$

The frequency-dependent load impedance Z_L from Equation (3) can be described by a simple zero-pole model as seen in Equation (1). The equation yields the $0 < A < B$ constraint. The parameters must be positive so that the stability is assured, and the $A < B$ constraint assures that the impedance has a low-pass characteristic, which is a trait of the cardiovascular system, and so a physiological constraint.

Substituting Equation (1) in Equation (3), we get the reflection coefficient described with the A and B parameters:

$$\begin{aligned} \Gamma(j\omega) &= \frac{Z_L(j\omega) - Z_c}{Z_L(j\omega) + Z_c} = \frac{B - A}{A + B + 2j\omega} = \\ &\frac{B - A}{2} \frac{1}{j\omega + \frac{B+A}{2}} = \frac{B - A}{B + A} \frac{1}{1 + \frac{j\omega}{\frac{B+A}{2}}}. \end{aligned} \quad (10)$$

So the gain of the reflection coefficient is given as $\frac{B - A}{B + A}$, and $\frac{B + A}{2}$ is the cutoff frequency, as the reflection coefficient has a low-pass characteristic.

Combining Equations (9) and (10), we get Equation (11), which is the transfer function described with the A and B parameters in the frequency domain.

$$\frac{P_{ao}(j\omega)}{P_{fem}(j\omega)} = \frac{\left(\frac{A+B}{2} + j\omega \right) e^{j\omega PTT} + \frac{B-A}{2} e^{-j\omega PTT}}{B + j\omega}. \quad (11)$$

In our previous work [10], a benchmark test was conducted to assess the robustness of the developed parameter identification method. By solving Equation (2) with one specific parameter set ($A_x; B_x$) and a given femoral aortic pressure input, simulated central aortic pressure curves were created and used as an input of the parameter identification method instead of real measurements to examine the behaviour of only the identification method.

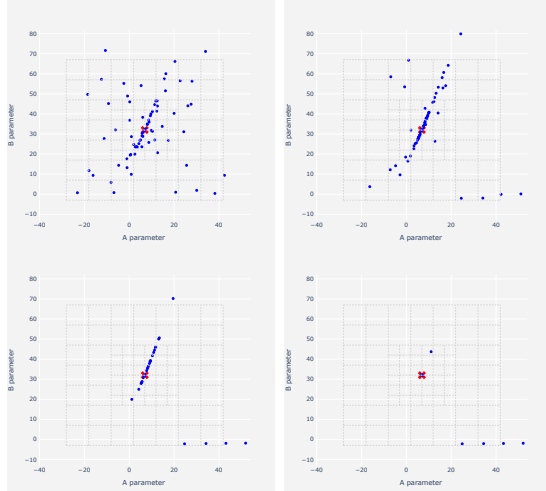


Fig. 2. The resulting parameter values after 4 (top left), 10 (top right), 20 (bottom left) and 40 (bottom right) training cycles during the benchmark test [10].

Figure 2 shows the changes of the A and B parameter values started from different initial values around the known optimal parameters ($A_x; B_x$) during the iterative optimization processes of the NeuralODE-based identification method. It can be seen in the Figure that the A and B parameters first converged to a line and then, following the line, moved to the known optimal parameters.

Examining Figure 2, there seems to be a linear relationship between the A and B parameters. The gain of the reflection coefficient described in 10 is proportional to the ratio of the two parameters mentioned above, so it could be used to describe the experienced linear relationship and, thus, be a better marker of the physiological state of the modelled blood vessels. Furthermore, physiological meaning can be associated with Γ , as well as the cutoff frequency ($\Theta = \frac{B+A}{2}$), which makes these better fit to describe physiological states during the different interventions on the subjects throughout the measurements.

Based on the Γ and Θ parameters, the A and B parameters can be exactly derived: $A = \Theta(1 - \Gamma)$, $B = \Theta(1 + \Gamma)$.

Using the inverse Laplace transform, Equation (9) can be converted to the time domain, and the differential equation that is used during the parameter identification process can be obtained using Γ and Θ instead of A, B:

$$\dot{P}_{ao,est}(t) + \Theta(1 + \Gamma)P_{ao,est} = \dot{P}_{fem}(t + PTT) + \Theta P_{fem}(t + PTT) + \Theta \Gamma P_{fem}(t - PTT) \quad (12)$$

In summary, the model can be equally described with the parameter set A, B as with Γ, Θ . As the latter parameter set has better physiological interpretation, so using Γ, Θ would better serve the analysis of the physiological state changes, which is one of our main research goals. Thus, Γ, Θ will be used for the system analysis.

D. Alternative Pulse Transit Time acquisition methods

Pulse Transit Time (PTT) is the time it takes the blood pressure wave to travel from the central arterial measurement site to the peripheral (femoral) arterial measurement site. It can be directly calculated using central arterial and femoral arterial blood pressure measurements, as the time delay between the two waves' feet represents the systole's initial point in each measurement site.

Using the measured pressure waves and calculating PTT for each heartbeat, the Γ and Θ parameters can be identified as seen in Figure 3 in the case of *pig8 base* section. The sampling rate of the measurement was 250 [Hz], meaning that the time resolution of the measurement is 0.004 [s], resulting in ± 0.004 [s] measurement error on the PTT calculation. Comparing this measurement error with the absolute value of PTT, having its maximum value in a given time period around 0.12 (see the top subfigure in Figure 3), the estimated relative error of the PTT calculation due to the 250 [Hz] sampling frequency is around 3.3%. This means that the inaccuracy of the calculated PTT caused by the relatively high sampling time has to be assessed - especially considering its effect on the identified parameters.

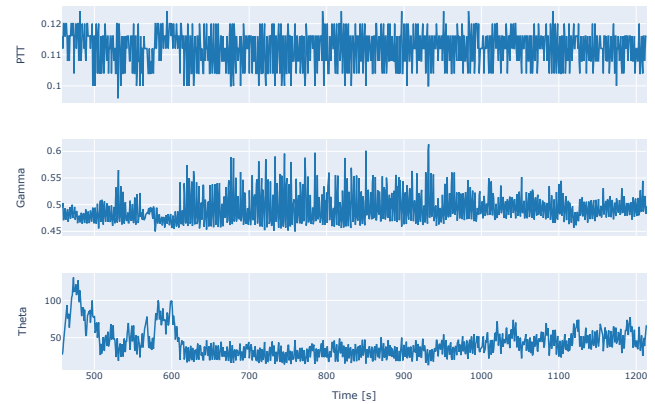


Fig. 3. The PTT calculated using the measurements (top), the identified Gamma and Theta parameters (middle and bottom) with relatively large variances likely due to the measurement noise caused inaccuracy of the calculated PTT parameter

Two alternative solutions are proposed and investigated to compensate for the excessive noise caused by inaccurate PTT measurement:

- 1) Assuming that the PTT is a slowly changing physiological parameter that can be considered constant in a few subsequent heartbeats and the time sampling is asynchronous to the heartbeats, then applying a moving

average on the calculated PTT values may be a way to compensate for the effect of measurement noise on the PTT.

- 2) Identifying PTT as a model parameter instead of calculating it based on measurement data.

Using a moving average to smooth the calculated PTT curve is an easy-to-implement solution and expectedly results in physiologically more accurate PTT values. Identifying the PTT as a model parameter has the potential advantage of better fitting simulated curves as this solution increases the order of freedom of the parameter identification by adding another parameter to be identified and minimising the curve fitting error.

III. RESULTS AND DISCUSSION

A. Alternative Pulse Transit Time acquisition methods

In Figure 4, the three ways of acquiring PTTs are shown in the case of the *pig8 base* section: the curve showing the PTTs calculated directly from the measurements, the curve of the averaged PTTs using a 7 sample moving window, and the curve of the identified PTT values.



Fig. 4. PTT values calculated directly from the measurements (blue), the curve of the averaged PTTs (orange) using a 7 sample moving window, and the curve of the identified PTT values (green) and the corresponding Γ parameters

The identified PTT values shown in Figure 4 significantly differ from the calculated ones, which results in different identified Γ parameters as well. This trend is consistent in other measurement sections of the experiments. Our goal is to examine the changes of physiological variable Γ through the different interventions. Thus, it is important that the Γ parameters are according to the subject's physiological state, which is ensured only if we do not identify but calculate the PTT values. However, using a moving average with a 7 sample window results in a smoother PTT curve and also in smoother identified Γ parameters. As a conclusion of these

Name	Mean	Std	Perc25	Median	Perc75
base	0.489	0.012	0.481	0.489	0.497
rm1	4.848	11.284	0.491	0.554	0.889
pause1	0.445	0.012	0.439	0.445	0.451
fluid1	0.469	0.014	0.46	0.468	0.477
pause2	0.41	0.02	0.395	0.405	0.429
rm2	8.451	15.935	0.379	0.44	10.947
pause3	0.386	0.012	0.381	0.387	0.393
endotoxin1	0.434	0.042	0.412	0.421	0.435
endotoxin_pause	39.78	55.4	0.534	1.38	96.987
endotoxin2	14.203	34.78	0.455	0.475	1.278
fluid2	-5.848	4.313	-8.737	-5.988	-1.016

TABLE I
STATISTICAL PROPERTIES OF Γ IN DIFFERENT SECTIONS OF THE MEASUREMENT OF *Pig 8*.

results the averaged PTT values are found beneficial for use in the parameter analysis studies.

B. Physiological parameter study of *Pig 8*

The measurement of *Pig 8* stopped soon after the endotoxin infusion. It started with a *base* sections, followed later by two *rm* sections, three *pause* sections, two *fluid* sections, and two *endotoxin* sections with a pause in between.

In the *base* section, the identified Γ values are consistently near 0.5, while Θ shows fluctuations. During this period, PTT values and the measured blood pressures are stable, as well as Γ . In the two *rm* sections, as the mechanical ventilation pressure rises, so does Γ , and Θ decreases to close to zero. In *pause1* and *fluid1* sections, Γ values are stable, with fluctuating identified Θ values. In *pause2*, Γ decreases gradually to 0.4. In the *pause3* section, the parameters are stable; however, in the second part of the *endotoxin1* intervention, the Γ values rise, and Θ decreases. By the end of *endotoxin1*, Γ goes to 0.65 from 0.4, and this trend continues in *endotoxin_pause* with extreme high Γ , and low Θ values. By the end of the pause, the Γ values are stabilized in the physiologically correct range, however, the second endotoxin infusion has the same effect as the first, and this leads to an extreme drop in blood pressure, and the heart function stops.

In summary, in the sections from *base* until *endotoxin1*, the Γ parameters take on physiologically valid values (with occasional extreme spikes). In Figure 6, the boxplots of Γ in these sections are shown. On the y axis, logarithmic scale is used, enabling the visualization of both the physiological and pathological (related to the *rm* sections) ranges. It is visible that *fluid1* resulted in a slight increase of the Γ parameter.

Table I contains the descriptive statistics of the sections of the measurement of *Pig 8*.

IV. CONCLUSIONS

The NeuralODE-base parameter identification method is found to be efficient in the analysis of physiological-state-induced parameter changes of the artery. The physiological interpretation of the results became easier by using the proposed set of transfer function parameters, Γ and Θ . By the presented physiological analysis, the applicability of the

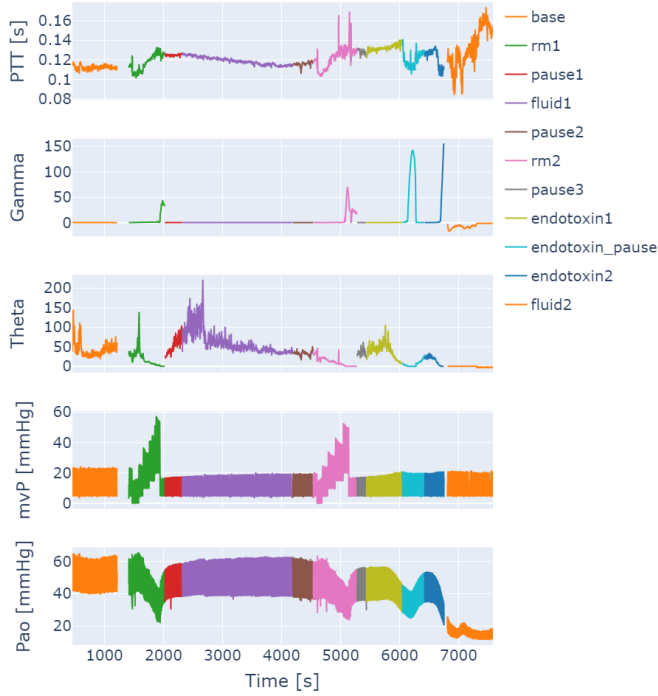


Fig. 5. The identified Γ and Θ parameters, the applied mechanical ventilation pressure (mvPw), and the arterial pressure (Pao) of Pig 8.

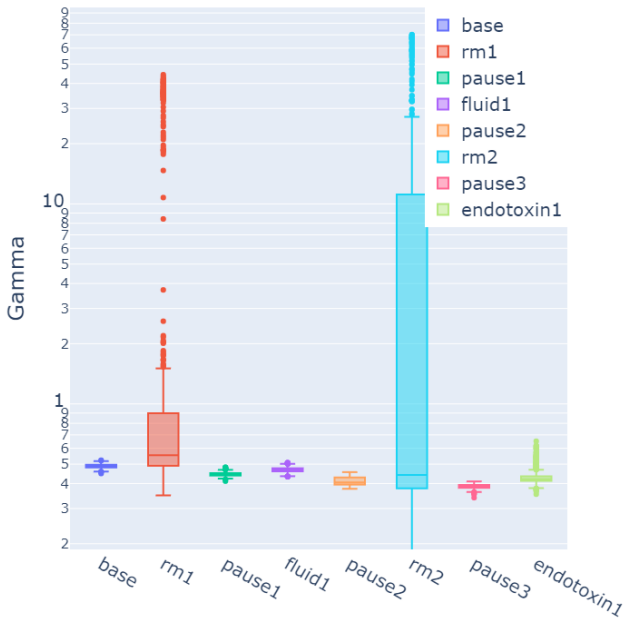


Fig. 6. The box plots of Γ parameters in sections of Pig 8.

proposed methodology is demonstrated. Using the methodology for the analysis of the available data sets will be a substantial help in improving our understanding of the effect of different clinical interventions and developing the model-

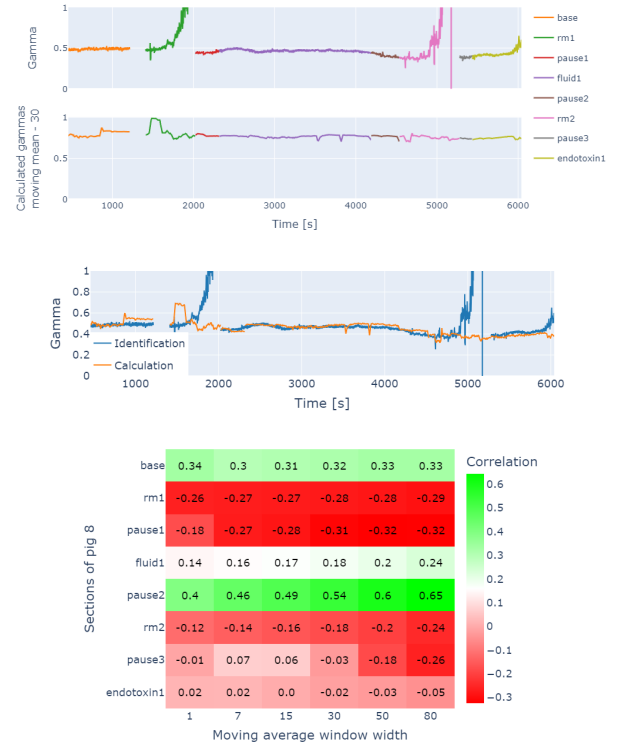


Fig. 7. The identified and calculated Γ values in different sections of Pig 8 in separate axes (top), superimposed with the means adjusted (middle), and the correlation values between the identified Γ curve and the moving averaged calculated Γ curves with different (1, 7, 15, 30, 50, 80) sample wide windows (bottom).

based biological marker identification method, enabling better control of fluid therapy.

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